Program : **Diploma in Information Technology**	
Course Code : **4267**	Course Title: **Computer System Administration Lab II**
Semester : **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To provide practical experience with the concept of System Administration tools.
- To implement the basic concept of managing software, RPM and YUM.
- To provide practical experience for managing accounts, disk and file system.

Course Pre-requisites:

Topic / Description	Course Code	Course Title	Semester
Basic knowledge of partitioning		Computer Hardware and Networking Lab	3
Basic concepts of Linux		System Administration Lab-I	3

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply concepts of system administration, administration tools and different partition types	12	Applying
CO2	Apply the basic concepts of managing software, RPM and YUM	12	Applying
CO3	Build the concepts of managing user account, create user account and create group account	12	Applying
CO4	Apply the concept of managing disk and file system	6	Applying

	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3	3					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Apply Concepts of System Administration, Administration Tools and different Partition Types		
M1.01	Experiment with a GNOME desktop, open the date and time window. Check the time zone is set properly.	1	Applying
M1.02	Make use of the system monitor to sort all processes running on your system by user name. Notice which user run which processes.	1	Applying
M1.03	Build all files under the /var/spool directory that are owned by users other than root and display a long listing of them.	2	Applying
M1.04	Develop Login as a regular user and become root using su. Edit the /etc/sudoer file to allow your regular user account to have full root privilege via the sudo command.	2	Applying
M1.05	Build installation of Fedora from live media, using as many of the default options as possible. After completing the installation ofFedora, update the entire package on the system.	2	Applying
M1.06	Experiment with installation from an RHEL installation DVD, make it so that installation runs in text mode.	2	Applying

M1.07	Build installation from an RHEL installation DVD and set the disk partitioning	2	Applying
CO2	Apply the Basic Concepts of Managing Software, RPM and YUM		
M2.01	Build Rpm Installation, upgradation , remove , reinstall	1	Applying
M2.02	Build Yum installation, upgradation , remove , reinstall	1	Applying
M2.03	Build Yum installation using RPM	1	Applying
M2.04	Make use of the YUM repository for the package that provides the mogrify commands	1	Applying
M2.05	Organize all the documentation files contained in the package that provides the mogrify command	2	Applying
M2.06	Experiment with List, Search, get information of package using YUM	2	Applying
M2.07	Build all available packages using YUM and all Installed packages using YUM	2	Applying
M2.08	Experiment with installing, Querying and verifying software with rpm command	2	Applying
	Lab Exam – I	1.5	
CO3	Build the Concepts of Managing User Account, Create User Account and Create Group Account		
M3.01	Build how Managing user account: a) Creating user and setting password b) Create user with default setting Specifies a user full name when creating a user	2	Applying
M3.02	Construct how adding a user with non-default home directory	2	Applying
M3.03	Experiment with how to Create a user while copying contents to the home directory	2	Applying
M3.04	Experiment withManaging group account a) Create group using group address command b) Creating a group with specified GID Adding a new user	2	Applying
M3.05	Build a named group account and test that it uses group ID.	2	Applying

M3.06	Experiment with how to Add jbaxter to testing group and the bin group	2	Applying
CO4	Apply the Concept of Managing Disk and File System.		
M4.01	Apply a command to list the partition table for the USB flash drive	1	Applying
M4.02	Build how to Add three partitions to the USB flash drive :100MB Linux partition, 200MB Linux partition and 500MB LVM partition. save changes	1	Applying
M4.03	Build swap partition and turn on so additional swap is immediately available	1	Applying
M4.04	Experiment with how to Grow the logical volume from 200MB to 300MB	2	Applying
M4.05	Experiment with how to Create mount point called /mnt/mypart, and mount Linux partition on it.	1	Applying
	Lab Exam – II	1.5	

*****Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do experiments as a group of 2-3. There is no duplication in experiments between groups. Experiments should include the concepts of system administration, system administration tool,RPM)

Example: Create a new regular user on the local system and

1. Open the /etc/passwd file. Find a current regular user with UID of 1000 or above
2. Try to logging in as the modified user.
3. Identify the default shell.

Text / Reference:

T/R	Book Title / Author
T1	Christopher Negus "Linux Bible The comprehensive tutorial resource" 9th edition

Online Resources:

Sl. No.	Website Link
1	https://www.tuxcademy.org/download/en/adm1/adm1-en-manual.pdf
2	https://www.tecmint.com/20-linux-yum-yellowdog-updater-modified-commands-for-package-mangement/

3	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/deployment_guide/s2-users-cl-tools